

Phase 4 – Project Planning Phase

PROJECT: EduGenie – Google Gemini Powered Learning Assistant

OBJECTIVE

To prepare a clear development plan, divide the work into milestones, and organize the project files and activities.

PROJECT MILESTONES

Milestone 1 – Model Selection and Architecture

- Select Gemini for Q&A, summarization, quiz generation, and learning paths.
- Select LaMini-Flan-T5-783M for concept explanation.
- Prepare the project folder architecture.

Milestone 2 – Core Functionalities

- Implement explanation module.
- Implement QnA module.
- Implement quiz module.
- Implement summary module.
- Implement learning path module.
- Create FastAPI endpoints.

Milestone 3 – Frontend Development

- Build HTML interface.
- Add CSS styling.
- Connect frontend with backend.

Milestone 4 – Deployment and Testing

- Run the application locally using Unicorn.
- Test every feature.
- Verify the final application.

PROJECT FOLDER STRUCTURE

EduGenie/

```
main.py
explanation_module.py
qna.py
quiz_module.py
summary_module.py
learning_path.py
templates/
    index.html
static/
    style.css
```

PROCEDURE

1 Divide the project into milestones.

2. Assign modules and development activities.
3. Prepare the folder structure.
4. Plan backend implementation.
5. Plan frontend implementation.
6. Plan integration and testing activities.